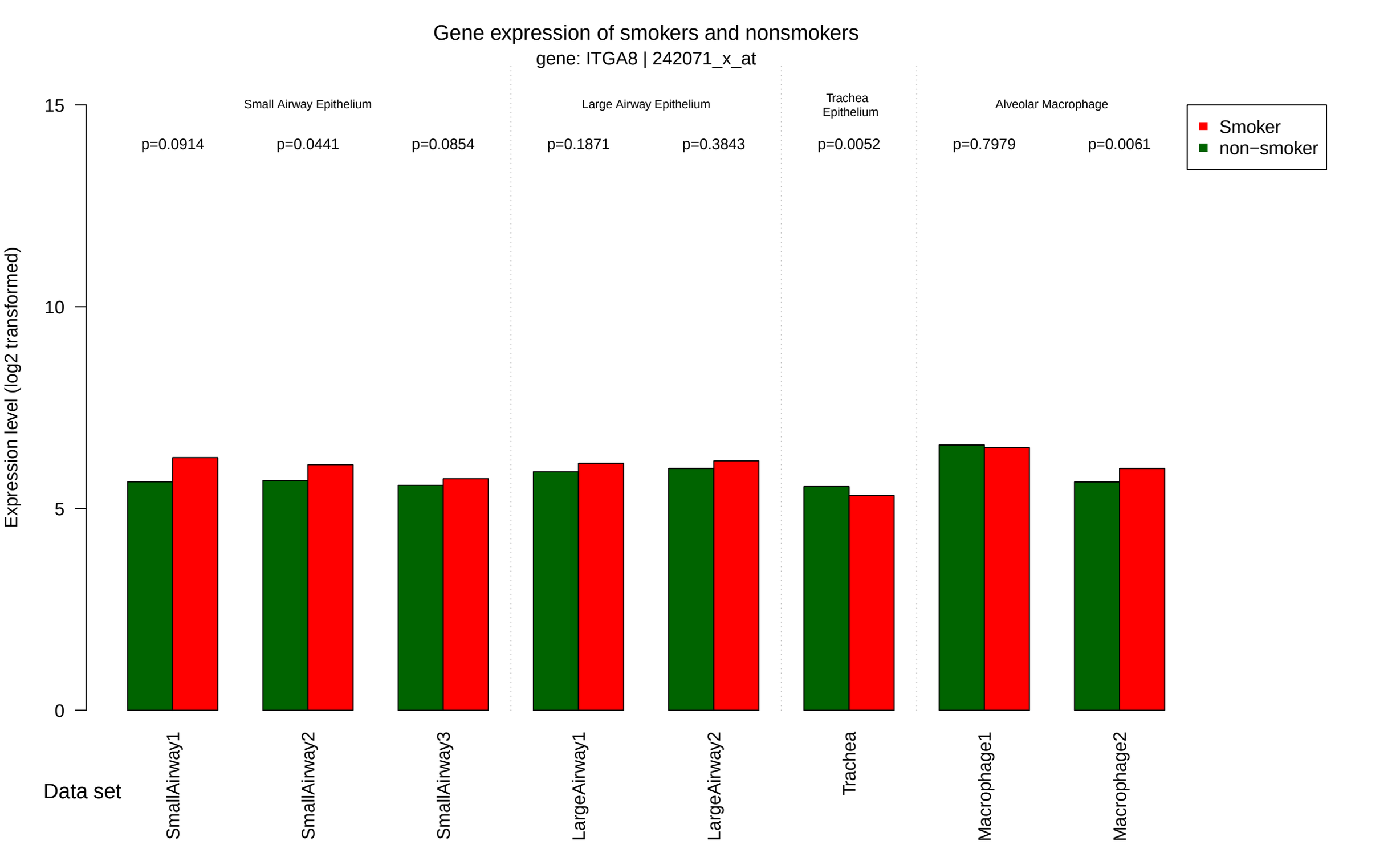
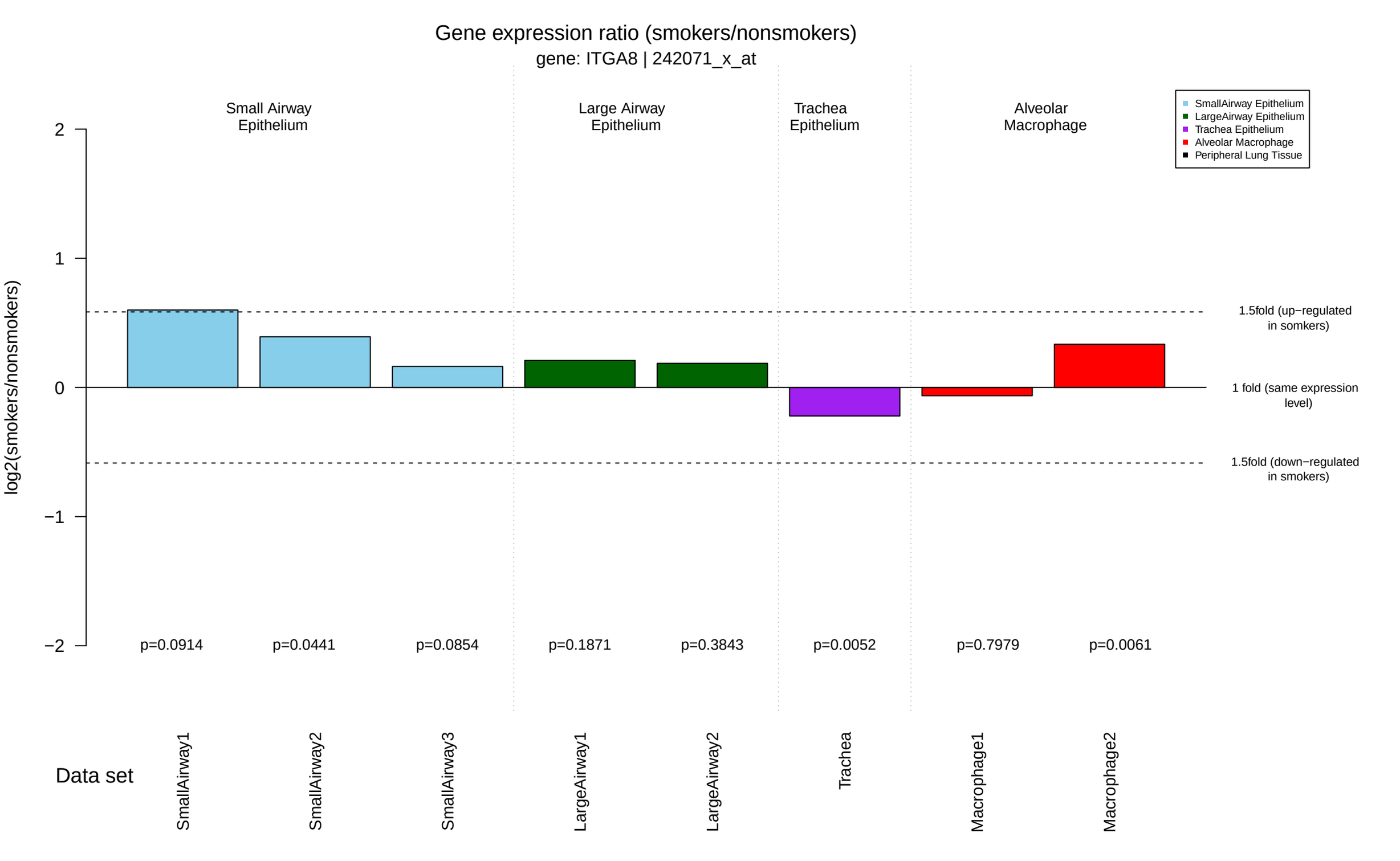
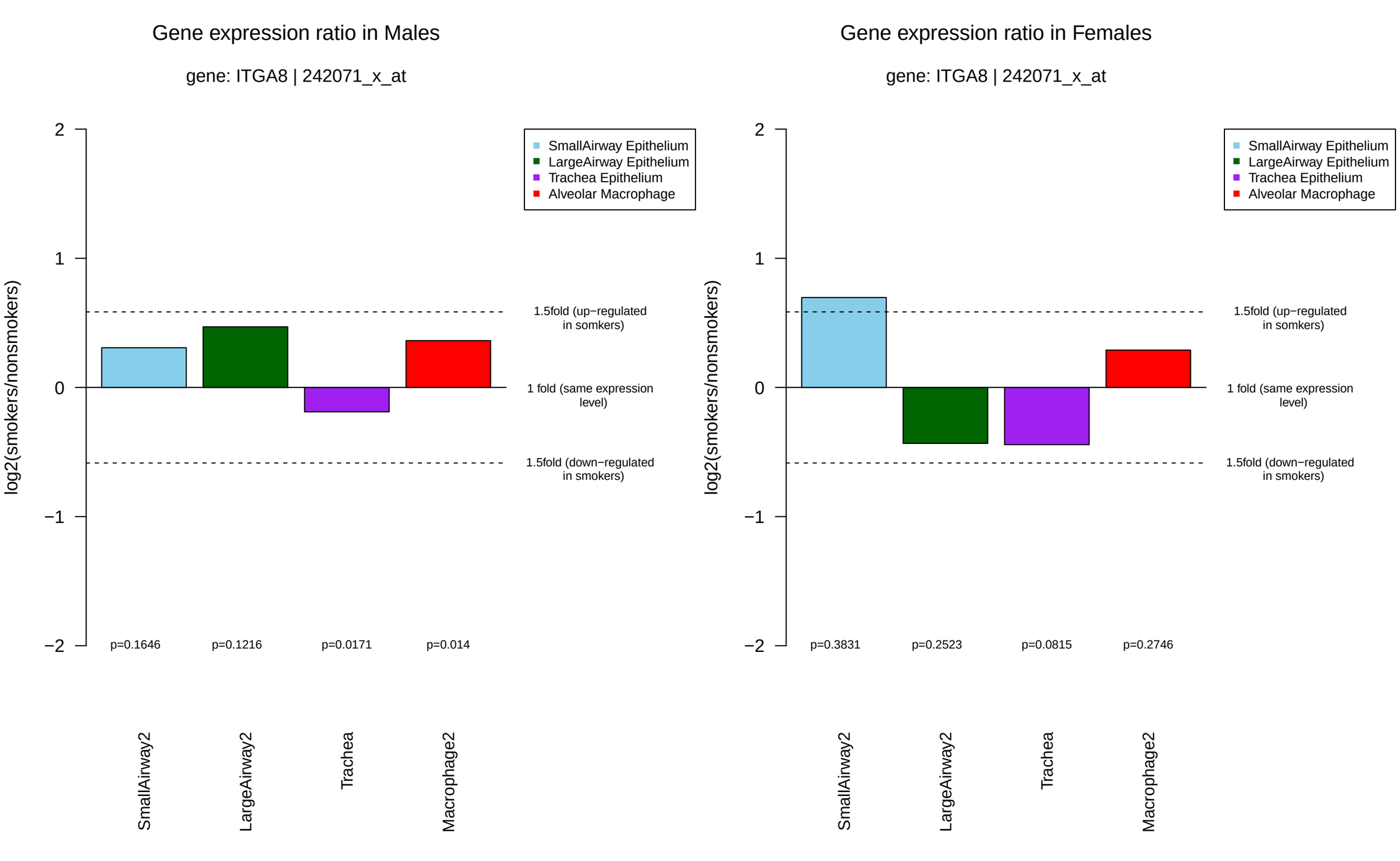
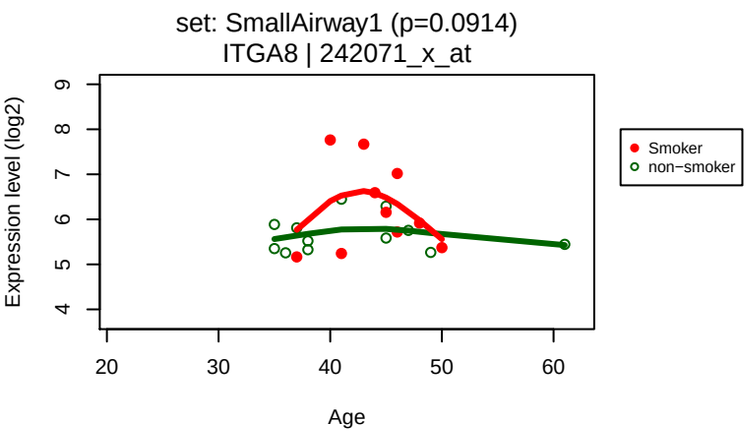
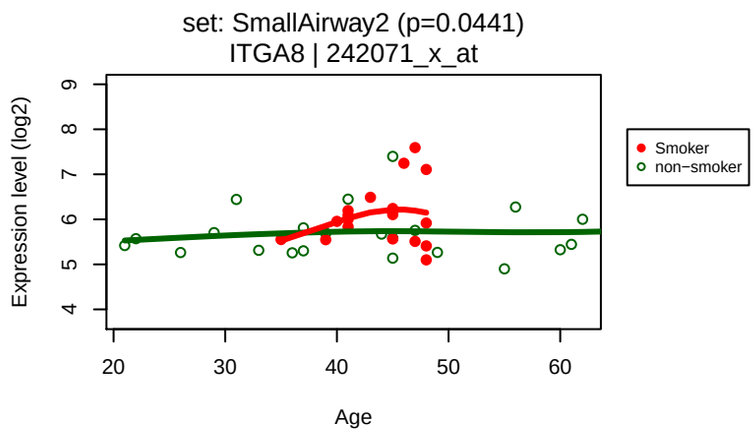
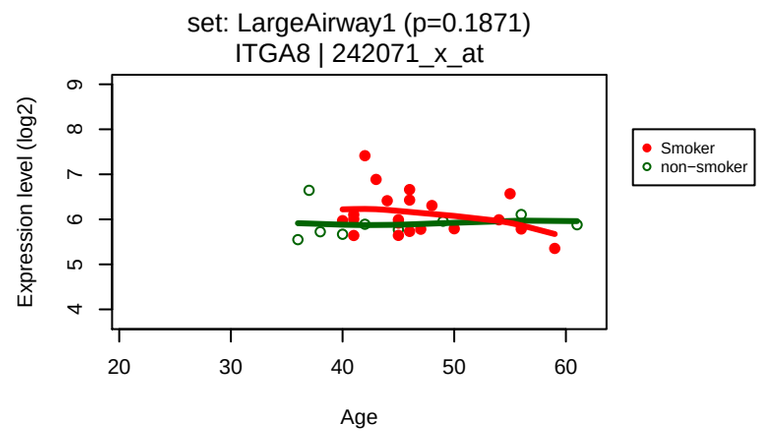

Gene expression of smokers and nonsmokers
gene: ITGA8 | 242071_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0914
p=0.0441
p=0.0854
p=0.1871
p=0.3843
p=0.0052
p=0.7979
p=0.0061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

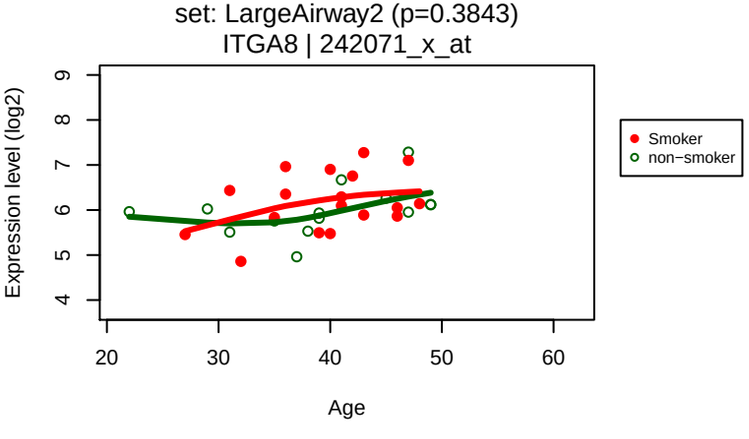

Gene expression ratio (smokers/nonsmokers)
gene: ITGA8 | 242071_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0914
p=0.0441
p=0.0854
p=0.1871
p=0.3843
p=0.0052
p=0.7979
p=0.0061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

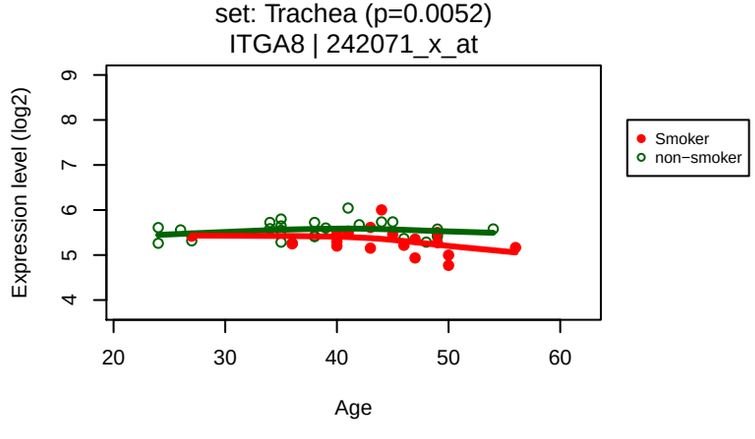

Gene expression ratio in Males
gene: ITGA8 | 242071_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1646
p=0.1216
p=0.0171
p=0.014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITGA8 | 242071_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3831
p=0.2523
p=0.0815
p=0.2746
SmallAirway2
LargeAirway2
Trachea
Macrophage2

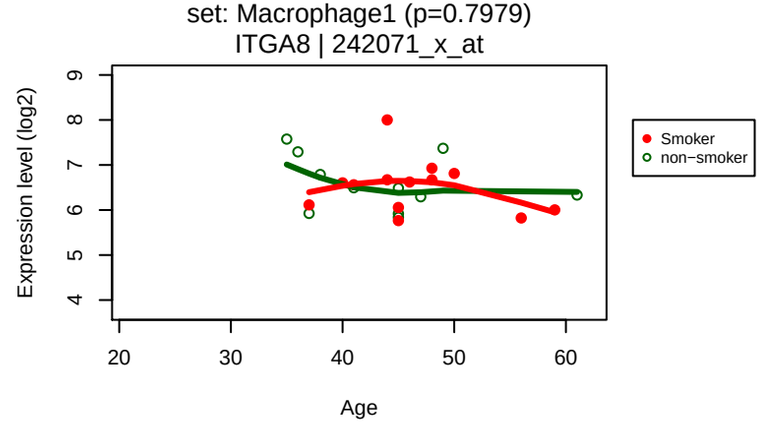

set: SmallAirway1 (p=0.0914)
ITGA8 | 242071_x_at
Expression level (log2)
Age
Smoker
non−smoker

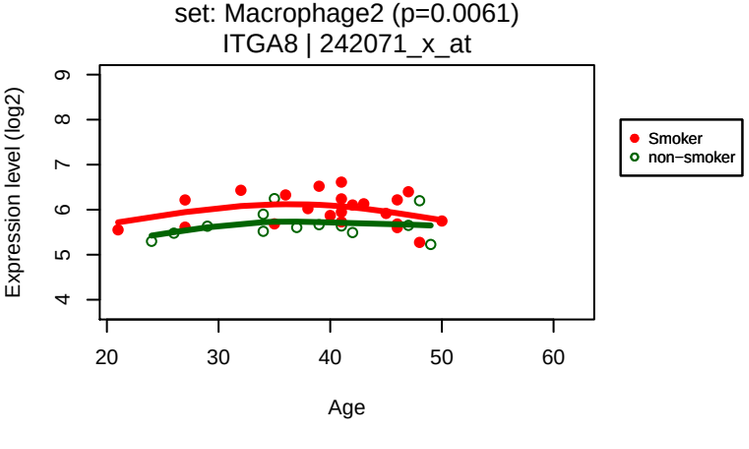

set: SmallAirway2 (p=0.0441)
ITGA8 | 242071_x_at
Expression level (log2)
Age
Smoker
non−smoker

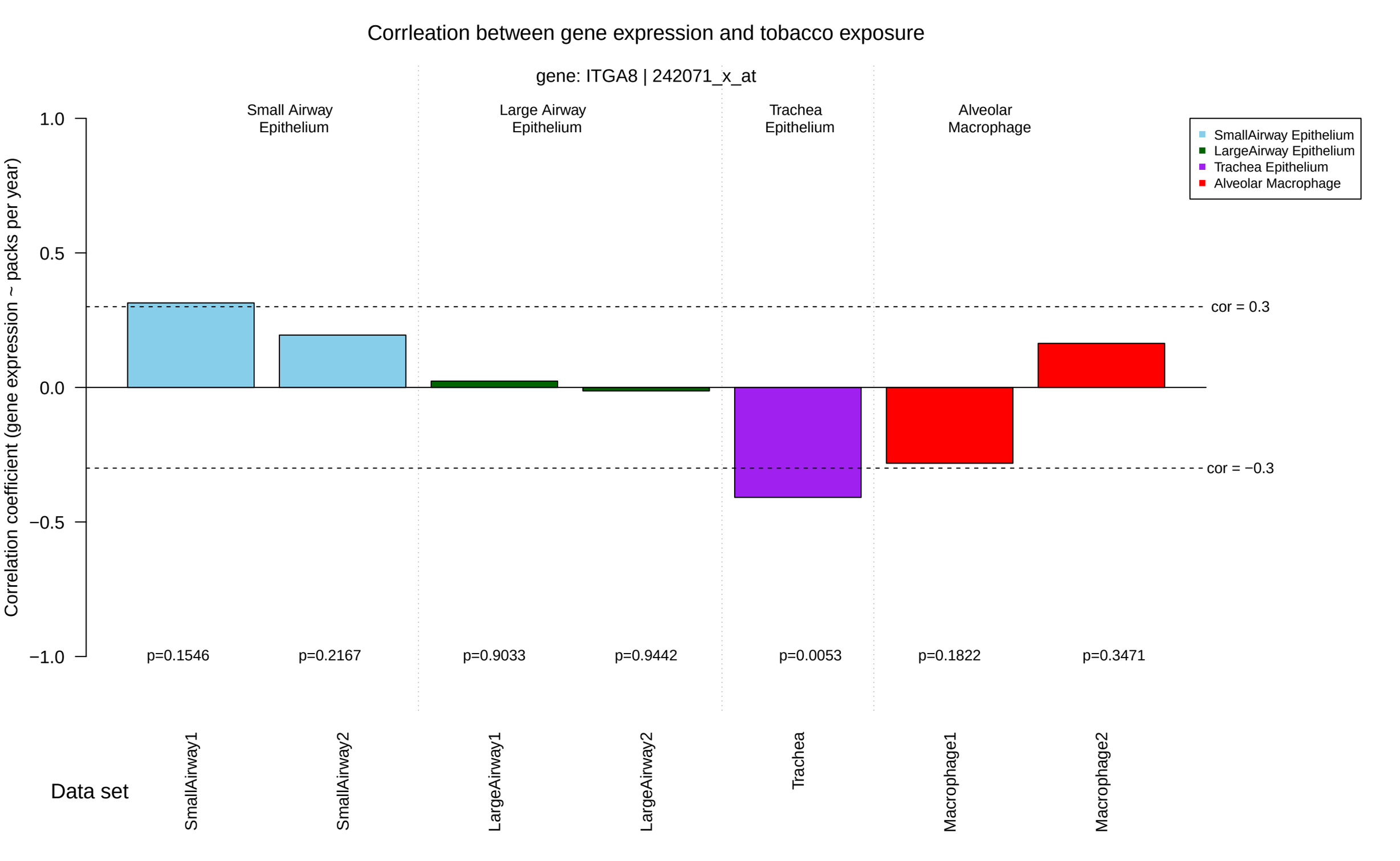

set: LargeAirway1 (p=0.1871)
ITGA8 | 242071_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3843)
ITGA8 | 242071_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0052)
ITGA8 | 242071_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7979)
ITGA8 | 242071_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0061)
ITGA8 | 242071_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITGA8 | 242071_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1546
p=0.2167
p=0.9033
p=0.9442
p=0.0053
p=0.1822
p=0.3471
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2